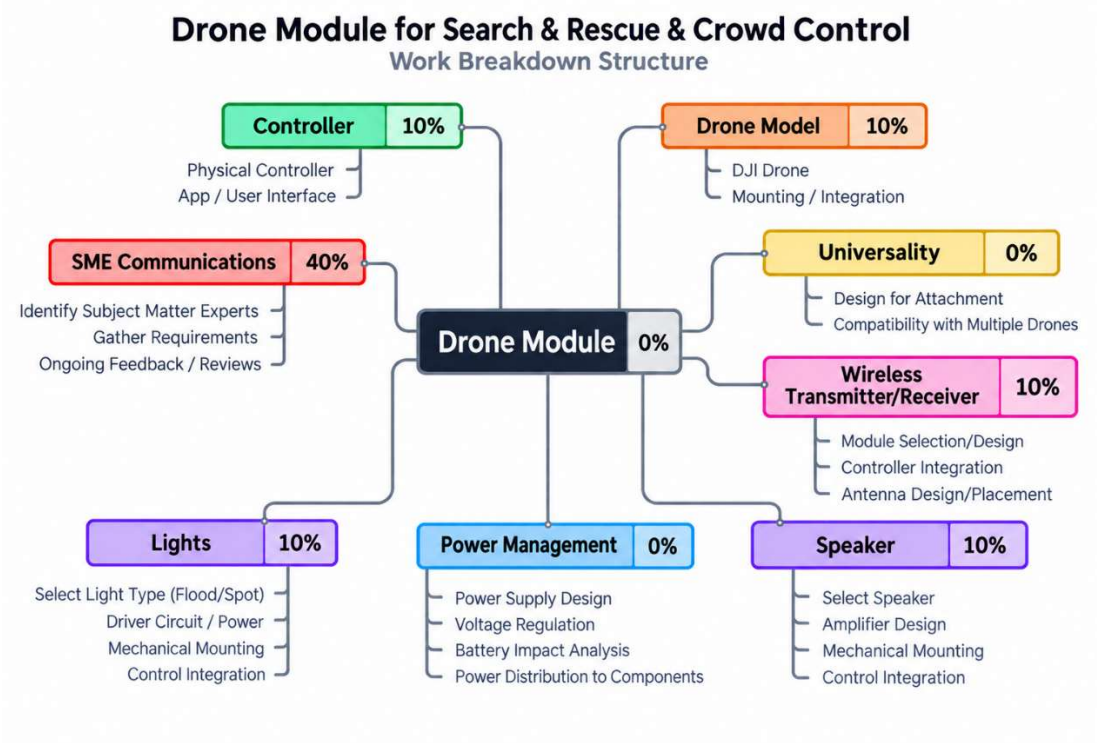
ELEC 4010 Capstone 1 Weekly Status Report

Project Name:	Drone Mounted Emergency Communications System
Team Members:	Hamilton Crews, Beau Blackburn, Tyler Engle, Joseph Rusk, Alejandro Matlacuatzi, Creed Allen
Week:	3
Date:	9/18/26
Website link:	https://josephrusk.github.io/team2-capstone-site/

Project description in plain language, approximately 25 words:
Design and development of a drone mounted communications module hosting speakers, cameras, and lights. Module is to be used by search and rescue and law enforcement agencies to reach people that are unable to communicate through other means.

TASK STATUS - VISUAL

(1) Draw the Work Breakdown Structure for your project



(2) Show the updated Gantt Chart

Capstone Design Project	Semester 1																
Task Description	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14	Break	**Chart Key - Name**	**Abrev**
Contact SME's	**- ALL**															Alejandro Matlalcuatzi	**AM**
Write Design Proposal	**- CA**															Beau Blackburn	**BB**
Create Website	**- JR**															Hamilton Crews	**HC**
Contact Companies	**- TE, BB, HC, AM**															Tyler Engle	**TE**
Assign Report Duties	**- CA**															Joseph Rusk	**JR**
Assign Research Duties		**- CA**														Creed Allen (Lead)	**CA**
Research Lighting Systems			**- HC**													All Group	**ALL**
Research Speaker System			**- AM**														
Decide on Drone Model			**- TE**														
Confirm Module Constraints			**- TE**														
Research Module Casing			**- JR**														
Research Wireless Transmit/Receive			**- CA**														
Project Sponsorship Communication				**- CA, TE**													
Research Module Controller			**- BB**														
Meet With SME's																	

Accomplishments since last status report: The group has organized a meeting with Dr. Mathew Kirchner on Monday, September 24th to discuss mentoring our project. We have begun research into the modules subsystems and plan to decide on a base drone model for design constraints by the end of week four. Communication with Dr. Robert Dean about mentoring our project is still underway. We have also contacted Dr. Hunter Burch to ask about contacts with AFRL and other research laboratories to pitch for a project sponsor.

Obstacles encountered since last status report: A decision has been made to include chassis connection to a specific drone model as part of the project. With this comes another decision about the design of the module casing, specifically whether or not the module is one piece or a collection of smaller sensor modules that can be selected for specific needs. This decision has expanded the scope of the project significantly and left the form of the final product unknown until further research.

Risks facing the project: Buying a drone without outside sponsorship could lead to a significant portion of the budget being spent prematurely before consideration of redesign costs. Group will have to make sure that any money to be spent in the second half of the course is well researched and absolutely necessary.

Objectives for the next week: Confirmation of SME support, continued research into module systems with a main goal of selecting a drone module for design constraints.

Comments: